

Kalman Filter — Multi-Step Ahead Forecasting

kalman_forecast.py — Mathematical Notes & Documentation

Growing Uncertainty Bands, Spread Forecast, Reversion Time Estimation

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Abstract. After the Kalman filter has processed all historical data, we hold the final state $\hat{\beta}_{T|T}$ and $P_{T|T}$. *Forecasting* extends this into the future: for the next h steps, we run only the *predict step* (no correction, since no new data exists yet), as stated in your Notion notes. The beta estimate stays constant ($\hat{\beta}_{T+h|T} = \hat{\beta}_{T|T}$) but uncertainty P grows linearly with h . This widening uncertainty band is the key output: it tells the trader how confident the model is about future spread values and estimates how long reversion will take.

Forecasting Without Observations

From your Notion notes:

Notion: Forecasting ($t' > t$)

For time $t' = t + 1, t + 2, \dots$ we simply run the prediction step *repeatedly without correction*, as we have no new market data yet.

$$S_{t'|t} = T_{t'-1} S_{t'-1|t}$$

$$\Omega_{s(t'|t)} = T_{t'-1} \Omega_{s(t'-1|t)} T_{t'-1}' + \Omega_{R_{t'} \eta_{t'}}$$

For our scalar pairs model ($T = 1$), this simplifies beautifully:

h -Step Beta Forecast

Formula 1 — Beta Forecast

$$\hat{\beta}_{T+h|T} = T^h \hat{\beta}_{T|T} = \hat{\beta}_{T|T} \quad (1)$$

Since $T = 1$ (random walk state), the best guess for β at any future time is simply today's estimate. The hedge ratio is assumed to not mean-revert; it follows a martingale.

Array form: $[\hat{\beta}_{T+1|T}, \dots, \hat{\beta}_{T+H|T}] = [\hat{\beta}_{T|T}, \dots, \hat{\beta}_{T|T}]$ (constant vector).

Growing Covariance $P_{T+h|T}$

Unlike the beta, the *uncertainty* about beta grows with each step into the future:

Formula 2 — h -Step Covariance

$$P_{T+1|T} = P_{T|T} + Q \quad (2)$$

$$P_{T+2|T} = P_{T+1|T} + Q = P_{T|T} + 2Q \quad (3)$$

$\vdots$

$$P_{T+h|T} = P_{T|T} + h \cdot Q \quad (4)$$

Uncertainty grows *linearly* with the forecast horizon h . A larger Q (process noise) means uncertainty grows faster.

Financial meaning. One day ahead: we're fairly sure about β . Thirty days ahead: much wider uncertainty. Eventually, $P_{T+h|T} \rightarrow \infty$ as $h \rightarrow \infty$, meaning we have no confidence about β far in the future.

Price Forecast $\hat{Y}_{T+h|T}$

Given future independent prices X_{T+h} (either known or estimated), the price observation forecast is:

Formula 3 — Price Observation Forecast

$$\hat{Y}_{T+h|T} = Z_{T+h} \hat{\beta}_{T+h|T} = X_{T+h} \cdot \hat{\beta}_{T|T} \quad (5)$$

The observation matrix $Z_{T+h} = X_{T+h}$ is the future independent price. We multiply it by our constant beta forecast.

Forecast Variance

The variance of the price forecast comes from two sources:

Formula 4 — Observation Forecast Variance

$$\begin{aligned} \text{Var}(\hat{Y}_{T+h|T}) &= Z_{T+h} P_{T+h|T} Z_{T+h}' + R \\ &= X_{T+h}^2 (P_{T|T} + hQ) + R \end{aligned} \quad (6)$$

- $X_{T+h}^2 P_{T+h|T}$: uncertainty about β multiplied by the price level (larger prices amplify uncertainty).
- R : irreducible measurement noise.

Both grow with h : the first grows via $P_{T+h|T}$; the second is constant.

Standard error:

$$\text{SE}(\hat{Y}_{T+h|T}) = \sqrt{X_{T+h}^2 (P_{T|T} + hQ) + R}$$

Spread Forecast

Formula 5 — Spread Forecast

$$\begin{aligned} \hat{S}_{T+h|T} &= \hat{Y}_{T+h|T} - \hat{\beta}_{T+h|T} \cdot X_{T+h} \\ &= X_{T+h} \cdot \hat{\beta}_{T|T} - \hat{\beta}_{T|T} \cdot X_{T+h} = 0 \end{aligned} \quad (7)$$

Key result: The expected spread forecast is *zero* under the random walk model. The spread is expected to remain at its long-run mean (zero if de-meaned).

This makes economic sense: if β doesn't change (random walk), the model predicts no systematic mispricing — the pair is expected to stay in equilibrium.

Why is this useful? The forecast itself is zero, but the *confidence interval* is not. The CI tells us:

- If the *current* spread is large (non-zero), when will it fall within $\pm 1.96\sigma_{T+h}$? (reversion horizon h^*)
- How wide is uncertainty at horizon h ? (position sizing)

Confidence Intervals

Formula 6 — Prediction Intervals

$$\begin{aligned} \text{CI}(h) &= \hat{Y}_{T+h|T} \pm z_{\alpha/2} \cdot \text{SE}_{T+h} \\ &= X_{T+h} \hat{\beta}_{T|T} \pm z_{\alpha/2} \sqrt{X_{T+h}^2 (P_{T|T} + hQ) + R} \end{aligned} \quad (8)$$

$$(9)$$

For 95% CI: $z_{0.025} = 1.96$.

Spread CI: Since $\hat{S}_{T+h|T} = 0$:

$$\text{CI}_{\text{spread}}(h) = \pm 1.96 \cdot \text{SE}_{T+h}$$

Expected Reversion Time h^* **Formula 7 — Reversion Time Estimate**

The current observed spread is S_T (last filtered value). The spread decays toward zero with approximate speed:

$$|S_{T+h}| \approx |S_T| \cdot e^{-\kappa h} \quad (10)$$

where $\kappa \approx -\ln |\hat{\rho}_1|$ and $\hat{\rho}_1$ is the lag-1 autocorrelation of past innovations ε_t .

Reversion horizon:

$$h^* = \left\lceil \frac{\ln |S_T / \mu_{\text{target}}|}{\kappa} \right\rceil \quad (11)$$

where $\mu_{\text{target}} = \mu_{\text{spread}} + \sigma_{\text{spread}}$.

Interpretation. h^* is the expected number of periods until the spread falls to within one standard deviation of its historical mean. This is used to size the holding period for the trade.

Worked Example**Setup**

Final filter state: $\hat{\beta}_{T|T} = 2.019$, $P_{T|T} = 3.88 \times 10^{-5}$
 $Q = 0.001$, $R = 0.01$, last $X_T = 51$.
 Forecast 3 steps ahead with $X_{T+h} = [51, 51.5, 52]$.

Step 1: Beta forecast (constant)

$$\hat{\beta}_{T+h|T} = 2.019 \quad \text{for all } h \in \{1, 2, 3\}$$

Step 2: Covariance growth

$$P_{T+1|T} = 3.88 \times 10^{-5} + 0.001 = 0.001039$$

$$P_{T+2|T} = 3.88 \times 10^{-5} + 0.002 = 0.002039$$

$$P_{T+3|T} = 3.88 \times 10^{-5} + 0.003 = 0.003039$$

Step 3: Price forecast

$$\hat{Y}_{T+1} = 2.019 \times 51 = 102.97$$

$$\hat{Y}_{T+2} = 2.019 \times 51.5 = 103.98$$

$$\hat{Y}_{T+3} = 2.019 \times 52 = 104.99$$

Step 4: Forecast variance and CI

$$\text{Var}_{T+1} = 51^2 \times 0.001039 + 0.01 = 2.72 + 0.01 = 2.73$$

$$\text{SE}_{T+1} = \sqrt{2.73} \approx 1.652$$

$$\text{CI}_{T+1} : 102.97 \pm 1.96 \times 1.652 = [99.7, 106.2]$$

Interpretation

The 95% CI widens from ± 3.24 at $h = 1$ to ± 5.6 at $h = 3$. This shows the growing uncertainty: the further we forecast, the less we trust our price estimate. For position sizing: at $h = 3$, if the current spread is within the CI, we cannot be confident it has departed from equilibrium — no trade.

Mean-Reverting Beta Model Extension

In the standard random walk model ($T = 1$), we assume the hedge ratio has no drift or anchor. However, in many financial markets, the hedge ratio itself may fluctuate around a long-run historical equilibrium $\bar{\beta}$. We can model this using an AR(1) state equation:

$$\beta_t - \bar{\beta} = \phi(\beta_{t-1} - \bar{\beta}) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q) \quad (12)$$

where $\phi \in (0, 1)$ represents the speed of regression of the hedge ratio to its long-run mean.

Mean-Reverting h-Step Projections

Under the AR(1) state model, the h -step-ahead beta forecast is:

$$\hat{\beta}_{T+h|T} = \phi^h \hat{\beta}_{T|T} + (1 - \phi^h) \bar{\beta} \quad (13)$$

and the state error covariance propagates as:

$$P_{T+h|T} = \phi^{2h} P_{T|T} + Q \frac{1 - \phi^{2h}}{1 - \phi^2} \quad (14)$$

As the horizon $h \rightarrow \infty$:

$$\hat{\beta}_{T+h|T} \rightarrow \bar{\beta} \quad (15)$$

$$P_{T+h|T} \rightarrow \frac{Q}{1 - \phi^2} \quad (16)$$

This represents a crucial difference: under the mean-reverting model, the uncertainty about the hedge ratio does not grow to infinity. Instead, it is bounded by the asymptotic state variance $\frac{Q}{1 - \phi^2}$.

Kelly Criterion & Position Sizing

A primary application of spread forecasts is the optimization of position leverage. Let the spread forecast at time $T + h$ be $\hat{S}_{T+h|T}$ and its forecast variance be $V_{S,h} = \text{Var}(S_{T+h} | \mathcal{F}_T)$.

If we assume the trade is closed at horizon h , the optimal leverage fraction f^* under the Kelly criterion is:

Dynamic Kelly Leverage

$$f_h^* = \frac{\mathbb{E}[S_{T+h|T}] - c}{V_{S,h}} \quad (17)$$

where c represents the transaction costs (spread crossings, brokerage, borrow fees) expressed in spread units, and:

$$V_{S,h} = X_{T+h}^2 \left(\phi^{2h} P_{T|T} + Q \frac{1 - \phi^{2h}}{1 - \phi^2} \right) + R$$

This highlights the importance of the growing uncertainty: as h increases, $V_{S,h}$ grows, which automatically scales down the optimal leverage f_h^* . This acts as a

risk-management mechanism, preventing the trader from over-leveraging long-horizon trades.

Code Flowchart — KalmanForecaster

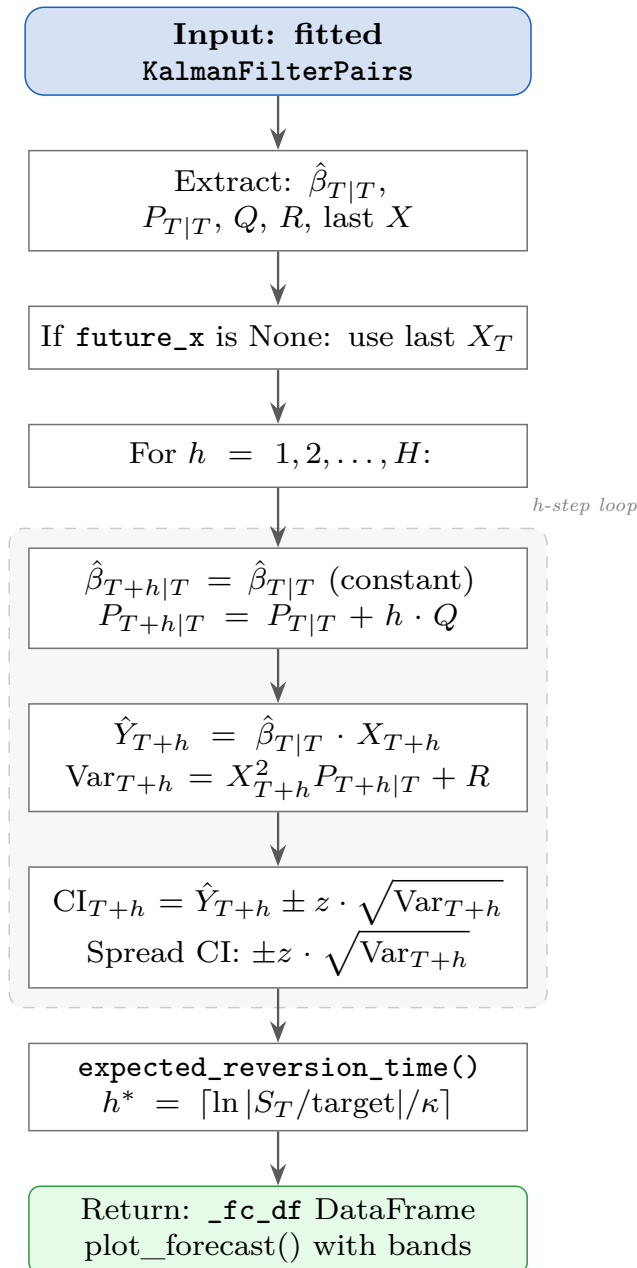


Figure 1: Execution flowchart for KalmanForecaster.forecast().

Python Implementation

forecast_beta and forecast_P

```

def forecast_beta(self, steps: int) -> np.ndarray:
    # Formula 1: beta constant under random walk
    return np.full(steps, self._beta_T)

def forecast_P(self, steps: int) -> np.ndarray:
    # Formula 2: P grows linearly with h
    h_arr = np.arange(1, steps + 1)
    return self._P_T + h_arr * self._Q
  
```

forecast() — main method

```

def forecast(self, steps=20, future_x=None):
    if future_x is None:
        future_x = np.full(steps, self._last_x)
    x_ext = self._extend_x(future_x, steps)
    beta_fc = self.forecast_beta(steps) # constant
    P_fc = self.forecast_P(steps) # growing
    y_fc = beta_fc * x_ext # Formula 3
    y_var = x_ext**2 * P_fc + self._R # Formula 4
    y_se = np.sqrt(y_var)
    spread_upper = self._z_crit * y_se # Formula 6
    spread_lower = -self._z_crit * y_se
    # Build DataFrame and return
  
```

expected_reversion_time()

```

def expected_reversion_time(self):
    innovations = np.array([s.innovation
                           for s in self._kf.states])
    if len(innovations) > 2:
        # Estimate kappa from innovation AR(1)
        acf1 = np.corrcorr(innovations[:-1],
                           innovations[1:])[0,1]
        if -1 < acf1 < 1 and acf1 < 0:
            kappa = -np.log(abs(acf1))
            h_star = ceil(
                log(|S_T|/target) / kappa)
    return max(1, min(h_star, steps))
  
```

Filter vs. Forecast: Key Distinction

Property	Filter ($t \leq T$)	Forecast ($t > T$)
Data used	$Y_1, \dots, Y_t$	$Y_1, \dots, Y_T$ (no new)
Step run	Predict + Update	Predict only
$\hat{\beta}$ changes	Yes (per update)	No (constant)
P behaviour	Decreases on update	Increases by Q
Spread	$Y_t - \hat{\beta}_{t t} X_t$	Zero (expected)
Use case	Live trading	Position sizing, risk

Practical Guide: Using Forecasts for Trading

When to Act on Forecasts

- Entry check:** If current spread $|S_T|$ is large and $h^* < \text{max hold period}$, the trade is viable.
- Exit planning:** At $h = h^*$, the spread is expected to reach near-zero. Plan to close the position by then.
- Position sizing:** The CI width at h^* determines risk. Narrow CI $\rightarrow$ larger position; wide CI $\rightarrow$ smaller.
- Stop-loss:** If the forecast CI at $h = 1$ exceeds the stop-loss spread level, do not enter the trade.

Pipeline Position

Step	Module	Output
6	kalman_filter.py	$\hat{\beta}_{T T}, P_{T T}$
6.1	smoother.py	$\hat{\beta}_{t T}$ (offline)
6.2	dynamic_beta.py	Full pipeline
7	kalman_forecast.py	CI bands, h^*
7b	vecm_forecast.py	Price level + spread
8	signal/pair_signal.py	Entry/exit signals
9	backtest/engine.py	Strategy backtest

Interview Preparation

Key Interview Questions

1. **Why is the forecast spread zero in expectation?**

Because $\hat{\beta}_{T+h|T} = \hat{\beta}_{T|T}$ (random walk): the forecasted Y and the hedged X exactly cancel. The model believes the pair is in equilibrium going forward.

2. **Why does uncertainty grow linearly?**

Each step adds Q to P : $P_{T+h|T} = P_{T|T} + hQ$. Q represents the per-step variance of the β random walk.

3. **How does the CI help with position sizing?**

Wider CI $\rightarrow$ more uncertainty $\rightarrow$ smaller position. Kelly-style sizing: $f \propto \hat{S}_T^2 / \text{Var}_{T+h^*}$.

4. **What if future X is unknown?**

The code uses the last observed X_T as a flat projection. In practice, one can use VECM forecasts of X_{T+h} (from `vecm_forecast.py`).

5. **How is κ estimated from innovations?**

AR(1) of innovations: $\varepsilon_t = \rho\varepsilon_{t-1} + \xi_t$. $\kappa \approx -\ln |\hat{\rho}_1|$. This links filter residuals to OU speed.